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Electric linear actuator with anti-backdrive mechanism

Abstract

An electric linear actuator includes a housing extending along a longitudinal axis, the housing configured to be coupled to a first component. A spindle is provided in the housing and extends along the longitudinal axis from a proximal end to a distal end. A motor is coupled to the proximal end of the spindle and is configured to rotate the spindle. A spindle nut is received on the spindle and is movable with respect to the housing along the longitudinal axis in response to rotation of the spindle. The spindle nut is configured to be coupled to a second component. A one-way bearing prevents back-drive of the motor when an external load is applied to the second component while the motor is not provided with electrical power.

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Background/Summary

FIELD

(1) The present disclosure relates to electric linear actuators, such as for example actuators for trim tabs on boats.

BACKGROUND

(2) U.S. patent application Ser. No. 17/716,542, filed Apr. 8, 2022, which is hereby incorporated by reference herein, discloses an electric linear actuator having a housing extending along a longitudinal axis. A spindle is provided in the housing and extends along the longitudinal axis. A rod is coaxially coupled to the spindle and extends from a proximal end located in the housing to a distal end located outside the housing. The rod is movable along the longitudinal axis in response to rotation of the spindle. A circuit board is located in the housing adjacent the proximal end of the rod. The circuit board includes a linear inductive sensor having a linear axis oriented parallel to the longitudinal axis. The linear inductive sensor senses an actual position of the proximal end of the rod.

SUMMARY

(3) This Summary is provided to introduce a selection of concepts that are further described below in the Detailed Description. This Summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

(4) According to one example of the present disclosure, an electric linear actuator comprises a housing extending along a longitudinal axis. A spindle is provided in the housing and extends along the longitudinal axis from a proximal end to a distal end. A motor is coupled to the proximal end of the spindle and configured to rotate the spindle. A rod is coaxially coupled to the spindle and movable with respect to the housing along the longitudinal axis in response to rotation of the spindle. A one-way bearing is configured to prevent back-drive of the motor when the spindle is subject to an external load via the rod.

(5) According to some aspects, the electric linear actuator further comprises a thrust bearing provided coaxially on the spindle. The thrust bearing is located between the proximal end of the spindle and the rod.

(6) According to some aspects, the electric linear actuator further comprises a first bearing surface located on a proximal side of the thrust bearing and a second bearing surface located on a distal side of the thrust bearing.

(7) According to some aspects, one of the first and second bearing surfaces is provided on a thrust race that is provided coaxially on the spindle. The thrust race also serves as an inner race for the one-way bearing.

(8) According to some aspects, the other of the first and second bearing surfaces is provided on a thrust washer.

(9) According to some aspects, the spindle is configured to rotate within the thrust race as the motor rotates the spindle in a direction that retracts the rod into the housing.

(10) According to some aspects, the thrust race comprises a hollow cylindrical portion serving as the inner race for the one-way bearing and a flange extending radially outwardly from the cylindrical portion, the flange comprising the one of the first and second bearing surfaces.

(11) According to some aspects, the one-way bearing is located adjacent the proximal end of the spindle. According to some aspects, the one-way bearing is located between the proximal end of the spindle and the thrust bearing.

(12) According to some aspects, the one-way bearing is a roller bearing having rollers with axes of rotation parallel to the longitudinal axis.

(13) According to another example of the present disclosure, an electric linear actuator comprises a housing extending along a longitudinal axis, the housing configured to be coupled to a first component. A spindle is provided in the housing and extends along the longitudinal axis from a

proximal end to a distal end. A motor is coupled to the proximal end of the spindle and is configured to rotate the spindle. A spindle nut is received on the spindle and is movable with respect to the housing along the longitudinal axis in response to rotation of the spindle. The spindle nut is configured to be coupled to a second component. A one-way bearing is configured to prevent back-drive of the motor when an external load is applied to the second component while the motor is not provided with electrical power.

(14) According to some aspects, the electric linear actuator further comprises a thrust bearing provided coaxially on the spindle.

(15) According to some aspects, the electric linear actuator further comprises a thrust race provided coaxially on the spindle. The thrust race comprises both a radially oriented bearing surface for the one-way bearing and a longitudinally oriented bearing surface for the thrust bearing. According to some aspects, the thrust race comprises a hollow cylindrical portion comprising the radially oriented bearing surface and a flange extending radially outwardly from the cylindrical portion, the flange comprising the longitudinally oriented bearing surface.

(16) According to some aspects, the one-way bearing is configured to allow the thrust race and the spindle to rotate freely in a high-efficiency direction of the electric linear actuator. The spindle is configured to rotate within the thrust race as the motor rotates the spindle in an opposite, low-efficiency direction.

(17) According to some aspects, the one-way bearing is located adjacent the proximal end of the spindle. According to some aspects, the one-way bearing is located between the proximal end of the spindle and the thrust bearing.

(18) According to some aspects, the one-way bearing is a roller bearing having rollers with axes of rotation parallel to the longitudinal axis.

(19) According to some aspects, the electric linear actuator further comprises a rod configured to couple the spindle nut to the second component and configured to extend and retract from the housing as the spindle nut travels along the spindle.

(20) According to some aspects, the first component is a transom of a boat and the second component is a trim tab.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is described with reference to the following Figures. The same numbers are used throughout the Figures to reference like features and like components.

(2) FIG. 1 is a schematic of one example of an electric linear actuator according to the present disclosure, in which a motor is geared to a spindle of the actuator.

(3) FIG. 2 illustrates another example of an electric linear actuator according to the present disclosure, in which a motor directly drives a spindle of the actuator.

(4) FIG. 3 illustrates a trim tab coupled to a transom of a boat by way of an electric linear actuator like that shown in FIG. 1 or FIG. 2.

(5) FIG. 4 is a cross-section of a portion of the electric linear actuator of FIG. 2, taken along the line 4-4.

(6) FIG. 5 is a more detailed view of a portion of FIG. 4.

(7) FIG. 6 is an exploded view of parts of the electric linear actuator of FIG. 2.

(8) FIG. 7 is a schematic showing how parts of the electric linear actuator interact when power is provided to the motor to drive the spindle and a compressive load is applied to a rod coupled to the spindle.

(9) FIG. 8 is a schematic showing how parts of the electric linear actuator interact when the motor is no longer provided with power and a compressive load is applied to the rod.

DETAILED DESCRIPTION

(10) Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

(11) Unless otherwise specified or limited, the phrases “at least one of A, B, and C,” “one or more of A, B, and C,” and the like, are meant to indicate A, or B, or C, or any combination of A, B, and/or C, including combinations with multiple instances of A, B, and/or C. Likewise, unless otherwise specified or limited, the terms “mounted,” “connected,” “linked,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings. Further, unless otherwise specified or limited, “connected” and “coupled” are not restricted to physical or mechanical connections or couplings.

(12) As used herein, unless otherwise limited or defined, discussion of particular directions is provided by example only, with regard to particular embodiments or relevant illustrations. For example, discussion of “top,” “bottom,” “front,” “back,” “left,” “right,” “lateral” or “longitudinal” features is generally intended as a description only of the orientation of such features relative to a reference frame of a particular example or illustration. Correspondingly, for example, a “top” feature may sometimes be disposed below a “bottom” feature (and so on), in some arrangements or embodiments. Additionally, use of the words “first,” “second,” “third,” etc. is not intended to connote priority or importance, but merely to distinguish one of several similar elements from another.

(13) FIG. 1 is a schematic of one example of the innerworkings of an electric linear actuator **110** according to one example of the present disclosure. The electric linear actuator **110** includes a housing **112**, which extends along a longitudinal axis L. A spindle **114** is provided in the housing **112** and extends along the longitudinal axis L from a proximal end **136** to a distal end **138**. A rod **116** is coaxially coupled to the spindle **114** by way of a spindle nut **150** (e.g., a ball nut, lead nut, acme nut, etc.). The rod **116** extends from a proximal end **118** located in the housing **112** to a distal end **120** located outside the housing **112**. The rod **116** is movable along the longitudinal axis L in response to rotation of the spindle **114**, due to the coupling between the rod **116** and the spindle **114** provided by the spindle nut **150**.

(14) A motor **122** is configured to rotate the spindle **114**. The motor **122** is located in a motor housing **124** adjacent the housing **112**. The output shaft of the motor **122** is coupled to the input shaft of the spindle **114** by way of a gear train **126**. The motor **122** is provided with power from a power source **128**, such as a battery. A controller **130** is configured to control the motor **122**, such as by controlling the power provided to the motor **122** from the power source **128**. In one example, the controller **130** controls the power to the motor **122** using pulse-width modulation. Note that although the controller **130** and power source **128** are shown as being connected to the housing **112** of the electric linear actuator **110**, they are in fact connected to the electrical components inside the housings **112** and **124**, and not every signal or power connection is shown herein.

(15) The controller **130** includes a processing system **132**, a storage system **134** accessible by the processing system **132**, and an input/output (I/O) interface **135**, which relays information to and from the processing system **132**. The processing system **132** can comprise a microprocessor, including a control unit and a processing unit, and other circuitry, such as semiconductor hardware logic, that retrieves and executes software from the storage system **134**. The storage system **134** can comprise any storage media readable by the processing system **132** and capable of storing software. The storage system **34** can include volatile and non-volatile, removable and non-

removable media implemented in any method or technology for storage of information, such as computer-readable instructions, data structures, software modules, or other data. The processing system **132** loads and executes software from the storage system **134**, such as software programmed with a rod positioning algorithm, which direct the processing system **132** to operate to move the rod **116** to a setpoint position. For example, the controller **130** can be programmed to activate the motor **122** with a given voltage and/or current for a given time in order to move the rod **116** from a present position to a setpoint position according to the rod positioning algorithm.

(16) FIG. 2 illustrates another example of an electric linear actuator **10**, in which the motor **22** is in line with the longitudinal axis L along which the rod **16** travels. The electric linear actuator **10** includes a housing **12** extending along the longitudinal axis L. A spindle **14** is provided in the housing **12** and extends along the longitudinal axis L from a proximal end **36** to a distal end **38**. The spindle **14** shown here is a ball screw, but it could instead be a lead screw, a planetary roller screw, or an acme screw. The motor **22** is coupled to the proximal end **36** of the spindle **14** and is configured to rotate the spindle **14**. Such coupling could be made by way of a gear train or a direct-drive coupling. The motor **22** is a reversible motor that can rotate the spindle **14** in two opposite directions. A rod **16** is coaxially coupled to the spindle **14** and movable with respect to the housing **12** along the longitudinal axis L in response to rotation of the spindle **14**. The rod **16** has a proximal end **18** and a distal end **20**. Although not shown herein, the electric linear actuator **10** can be coupled to a controller and power source, which function as described hereinabove with respect to FIG. 1.

(17) The electric linear actuator **10** or **110** can be used to pivot a trim tab with respect to a transom of a boat. For example, referring to FIG. 3, a trim tab **40** and electric linear actuator **10**, **110** are shown installed on the transom **42** of a boat. Another trim tab may be installed on the opposite side of the transom **42**, which trim tab is aligned with the trim tab **40** and therefore hidden from view in FIG. 3. The electric linear actuator **10** or **110** (and thus the position of the trim tab **40**) is controlled by the controller **130** in response to inputs to a user input device, which may be a button, switch, key, or touch-sensitive display screen with a virtual key. A user may input a “trim up” command via the user input device, in response to which the controller **130** provides power to the electric linear actuator **10** or **110** to retract the rod **16** or **116** into the housing **12** or **112** by rotating the spindle **14** or **114** in a first direction (e.g., clockwise). The trim tab **40** can be trimmed up to the maximum trimmed up position shown at **40'**. The user may input a “trim down” command via the user input device, in response to which the controller **130** provides power to the actuator **10** or **110** to extend the rod **16** or **116** from the housing **12** or **112** by rotating the spindle **14** or **114** in an opposite, second direction (e.g., counterclockwise). The trim tab **40** can be trimmed down to the maximum trimmed down position shown at **40''**. The distal end of the trim tab **40** thus moves along the arc A as the rod **16** or **116** is extended from or retracted into the housing **12** or **112**. Note that the example of clockwise rotation of the spindle for trimming up and counterclockwise rotation of the spindle for trimming down are exemplary only; depending on the design of the spindle and spindle nut, the opposite could be true.

(18) Referring to both FIGS. 1 and 2, in order to increase the efficiency of the electric linear actuator **10**, **110** when the rod **16**, **116** is pushing against or pulling on a load, a thrust bearing **44**, **144** is provided, which decreases the friction that the motor **22**, **122** must overcome to rotate the spindle **14**, **114** and thereby extend or retract the rod **16**, **116** against a load. Through research and development, the present inventors have realized that such increased efficiency works both ways; that is, increased efficiency also makes it easier for the spindle **14**, **114** to rotate in response to an external load being applied to the rod **16**, **116** while the motor **22**, **122** is not provided with power. Such rotation of the spindle **14**, **114** as a result of an external applied load can back-drive the motor **22**, **122**, which is an undesired effect. In order to prevent such back-drive, the present inventors have provided a one-way bearing **46**, **146** about the spindle **14**, **114**. The one-way bearing **46**, **146** is configured to prevent back-drive of the motor **22**, **122** when the spindle **14**, **114** is subject to an

external load via the rod **16**, **116**. The one-way bearing **46**, **146** is a simpler and lower cost alternative to a spring-loaded electromagnetic brake and a more efficient alternative to a wrap-spring brake, both of which are typically used to prevent back-drive.

(19) Although the electric linear actuator **110** of FIG. **1** is shown with the thrust bearing **144** and one-way bearing **146** provided as part of the assembly, only the assembly of FIG. **2** will be described in detail, it being understood that the thrust bearing **144** and one-way bearing **146** operate in a similar manner to the thrust bearing **44** and one-way bearing **46** described herein below. It does bear mention, however, that the one-way bearing **146** in the electric linear actuator **110** could be provided other than coaxially with the spindle **114** as shown herein, such as on a shaft in the gear train **126** or between the gear train **126** and the output shaft of the motor **122**.

(20) FIG. **2** shows the electric linear actuator **10** with the housing **12** partially cut away to reveal the inner components, while FIG. **4** shows a cross-section taken along the line **4-4** in FIG. **2**. The housing **12** is configured to be coupled to a first component, such as the transom **42** of a boat, by way of a clevis **48** and a mounting bracket (not shown). The spindle **14** is provided in the housing **12** and extends along the longitudinal axis **L** from a proximal end **36** to a distal end **38**. The motor **22** is coupled to the proximal end **36** of the spindle **14**, such as by way of a direct drive coupling **23**, and is configured to rotate the spindle **14** when power is provided to the motor **22**. The spindle nut **50** is received on the spindle **14** and movable with respect to the housing **12** along the longitudinal axis **L** in response to rotation of the spindle **14**. The spindle nut **50** is configured to be coupled to a second component, such as a trim tab **40**. More specifically, the rod **16** is configured to couple the spindle nut **50** to the second component (e.g., trim tab **40**) by way of a clevis **52** and a mounting bracket (not shown). The spindle nut **50** is positioned between upper and lower shoulders formed within the rod **16** and thus the rod **16** is configured to extend and retract from the housing **12** as the spindle nut **50** travels along the spindle **14**. Such extension and retraction of the spindle **14** moves the second component with respect to the first component, or vice versa. The one-way bearing **46** is configured to prevent back-drive of the motor **22** when an external load is applied to the second component (e.g., trim tab **40**) while the motor **22** is not provided with electrical power.

(21) FIG. **5** shows a close-up of the cross-sectional view of FIG. **4** and is referred to in order to describe the bearing assembly **45** of the electric linear actuator **10** in more detail. The bearing assembly **45** is situated within a bore **13** formed in the housing **12**. It can be seen that the thrust bearing **44** is provided coaxially on the spindle **14**, more specifically, on a lesser-diameter portion **55** of the spindle situated above the shoulder **54**. The thrust bearing **44** is located between the proximal end **36** of the spindle **14** and the rod **16**. In the example shown here, the one-way bearing **46** is located adjacent the proximal end **36** of the spindle **14**. Specifically, the one-way bearing **46** is located between the proximal end **36** of the spindle **14** and the thrust bearing **44**. In other examples, the one-way bearing can be provided elsewhere, for example, between the thrust bearing **44** and the shoulder **54** on the spindle **14** or proximate the distal end **38** of the spindle **14**. In such alternative arrangements, the inner diameter of the bore **13** in the housing **12** would be decreased in order that the one-way bearing could be press fit therein.

(22) A thrust race **56** is provided coaxially on the spindle **14**. Referring to FIGS. **5** and **6**, the thrust race **56** comprises both a radially oriented bearing surface **58** for the one-way bearing **46** and a longitudinally oriented bearing surface **62** for the thrust bearing **44**. The thrust race **56** comprises a hollow cylindrical portion **56a** comprising the radially oriented bearing surface **58** and a flange **56b** extending radially outwardly from the cylindrical portion **56a**, the flange **56b** comprising the longitudinally oriented bearing surface **62**.

(23) More generally, as is typical for a thrust bearing **44**, the bearing assembly **45** provides for a first bearing surface **60** located on a proximal side of the thrust bearing **44** and a second bearing surface **62** located on a distal side of the thrust bearing **44**. One of the first and second bearing surfaces **60**, **62** (here, the second bearing surface **62**) is provided on the thrust race **56**, which is provided coaxially on the spindle **14**. The other of the first and second bearing surfaces **60**, **62**

(here, the first bearing surface **60**) is provided on a thrust washer **64**. In another example, the thrust race **56** could be designed and located such that the flange **56b** provides the bearing surface on the proximal side of the thrust bearing **44**, while a thrust washer could be provided on the distal side of the thrust bearing **44**. Typically, a thrust bearing will have a thrust washer on each side. In contrast, in the present example, one of the thrust washers is replaced by the flange **56b** on the thrust race **56**, which thrust race **56** also serves as an inner race for the one-way bearing **46**. Specifically, the hollow cylindrical portion **56a** serves as the inner race for the one-way bearing **46**, and the flange **56b** comprises the one of the first and second bearing surfaces for the thrust bearing **44**.

(24) FIG. **6** is an exploded view of the spindle **14** and the bearing assembly **45** provided on the lesser-diameter portion **55** of the spindle **14**. A washer **66** is provided immediately adjacent the shoulder **54** on the spindle **14**. The thrust race **56** is provided adjacent the washer **66**, with the cylindrical portion **56a** facing away from the washer **66**. The thrust bearing **44** is next provided over the cylindrical portion **56a** of the thrust race **56** and adjacent the bearing surface **62** on the flange **56b**. The thrust washer **64** is provided on the opposite side of the thrust bearing **44** from the flange **56b**. Two optional spacers **68** are then provided over the cylindrical portion **56a** of the thrust race **56**. The spacers **68** may be needed for retrofit applications, in which the housing **12** is not of the appropriate dimension to hold the stacked components of the bearing assembly **45**. The one-way bearing **46** is next provided over the cylindrical portion **56a** of the thrust race **56**. Note that the one-way bearing **46** does not have an inner race, so the rollers **70** can be seen in the outer race **72**. The bearing surface **58** on the cylindrical portion **56a** of the thrust race **56** instead serves as the inner race, against which the rollers **70** rotate. Finally, a retaining ring **76** is provided on the lesser-diameter portion **55** of the spindle **14**.

(25) Note that in the present example, the one-way bearing **46** is a roller bearing having rollers **70** with axes of rotation parallel to the longitudinal axis **L**. However, in other examples, the rollers could be radially oriented perpendicular to the longitudinal axis **L** (as with the rollers of the thrust bearing **44**) and one of the upper and lower races of the one-way bearing could be keyed to the thrust race **56**, while the other of the upper and lower races could be keyed to the inside of the housing **12**.

(26) FIG. **7** shows the bearing assembly **45** in a case in which the rollers **70** of the one-way bearing **46** rotate freely and allow rotation of the thrust race **56** and spindle **14**, while FIG. **8** shows the bearing assembly **45** in a case in which the spindle **14** is rotated in such a way that the rollers **70** of the one-way bearing **46** are locked. FIGS. **7** and **8** also show how an optional bushing **78** can be provided to serve as a housing for the one-way bearing **46**, as well as how the spindle **14** can be driven by the last gear **80** in a gear train, in the event the spindle **14** is not directly driven by the motor **22**.

(27) In FIG. **7**, the motor **22** is rotating the spindle **14** in a direction in which the rollers **70** of the one-way bearing **46** rotate freely. In this example, such rotation of the spindle **14** results in the rod **16** extending from the housing **12**; however, in cases where the electric linear actuator **10** is typically used to retract the rod **16** into the housing **12** against a load, the one-way bearing may allow free rotation of the spindle in a direction tending to retract the rod. As the spindle **14** rotates, the thrust race **56** also rotates due to friction between the outer circumferential surface of the spindle **14** and the inner circumferential surface of the cylindrical portion **56a** of the thrust race **56**. The thrust race **56** rotates efficiently due to the thrust bearing **44** being adjacent the flange **56b** of the thrust race **56**. The thrust bearing **44** effectively lessens the friction against which the thrust race **56** (and therefore spindle **14**) would otherwise have to work. The spindle **14** therefore rotates efficiently, in this example, to extend the rod **16** from the housing **12**.

(28) A load may be applied in a direction opposite the direction of extension of the rod **16**. This is typically the case for a trim tab actuator, the motor of which must be able to extend or hold the rod **16** against loads as high as 750 pounds as the boat bounces and the trim tab slams into the water, or more typically around 300 pounds of sustained load at cruising. This compressive load is shown at

74 in FIG. 7. Because the one-way bearing **46** is not locked, the load transmitted from the rod **16** to the spindle **14** is thereafter transmitted through friction by way of the washer **66** into the thrust race **56**, then the thrust bearing **44**, then the thrust washer **64**, and from there into the housing **12**. If the thrust bearing **44** were not present, such compression loading would tend to increase friction and might make it difficult for the motor **22** to rotate the spindle **14**. However, the thrust bearing **44** reduces the friction, allowing the electric linear actuator **10** to operate at high efficiency to extend the rod **16**.

(29) Referring now to FIG. 8, if power is no longer provided to the motor **22** (e.g., power is lost or the trim system is turned off), there is no longer a torque applied to the spindle **14**. The compressive load **74** on the rod **16** will tend to rotate the spindle **14** in the absence of the counteracting torque applied by the motor **22**. The load on the spindle **14** will be transmitted through friction via the washer **66** to the thrust race **56**. However, the one-way bearing **46** prevents the thrust race **56** from rotating with the spindle **14**. The load is therefore transmitted from the thrust race **56** to the locked one-way bearing **46** and into the housing **12**, not into the motor **22**. Thus, the motor **22** is prevented from being back-driven when a compressive load **74** is applied to the rod **16** and the motor **22** is without power.

(30) Note that the motor **22** can still drive the spindle **14** to retract the rod **16** into the housing **12** despite the inclusion of the one-way bearing **46**. The motor **22** applies enough torque to rotate the spindle **14** within the cylindrical portion **56a** of the thrust race **56**. In other words, the motor's torque is enough to overcome the friction between the outer circumferential surface of the spindle **14** and the inner circumferential surface of the thrust race **56** such that the former can rotate within the latter. Thus, the spindle **14** is configured to rotate within the thrust race **56** as the motor **22** rotates the spindle **14** in a direction that retracts the rod **16** into the housing **12**.

(31) Those having ordinary skill in the art will understand that not all electric linear actuators are intended to push against an external compressive load, i.e., some actuators are intended to pull against an external tensile load. Thus, more generally, the one-way bearing **46** is configured to allow the thrust race **56** and the spindle **14** to rotate freely in a high-efficiency direction of the electric linear actuator **10**, i.e., as the spindle **14** rotates in a direction tending to extend the rod **16** against an external compressive load or to retract the rod **16** against an external tensile load. The spindle **14** is configured to rotate within the thrust race **56** as the motor **22** rotates the spindle **14** in an opposite, low-efficiency direction, i.e., as the motor **22** rotates the spindle **14** to retract a rod that typically works against an external compressive load or to extend a rod that typically works against an external tensile load. Thus, the bearing assembly **45** provides high efficiency in one direction (extension or retraction) of the rod **16** and low efficiency in the opposite direction of the rod **16**, while also preventing back-drive of the motor **22** when the motor **22** is not provided with power. In the example in which the electric linear actuator **10** is used to pivot a trim tab **40**, high rotational efficiency is desirable in the direction of extension of the rod **16**, as the motor **22** is working against an external compressive load, while low efficiency in the direction of retraction of the rod **16** is sufficient, as the compressive load acts in the direction of retraction.

(32) The load that the one-way bearing **46**, **146** needs to be able to handle is dependent upon not only the torque that is expected to act on the spindle **14**, **114** when an external load is applied to the rod **16**, **116**, but also on the type and design of the spindle **14**, **114**. For instance, if a ball screw is used, the one-way bearing will likely need to be able to handle a higher load than if an acme screw were used, all else being equal. A ball screw with a larger lead angle (and therefore higher efficiency) would likely need to be paired with a one-way bearing that can handle greater loads, while a ball screw with a smaller lead angle could be paired with a one-way bearing that can handle lesser loads, all else being equal. Friction within the assembly will also play a part in selection of the one-way bearing **46**, **146**. For instance, a lead screw has higher drag torque—and therefore lower efficiency—than a ball screw, and thus may be able to be paired with a one-way bearing that can handle less load, all else being equal.

(33) Although thrust bearing **44** shown herein is a roller bearing, it could instead be a tapered roller bearing or a ball bearing. The one-way bearing **46** could be a one-way needle roller bearing as shown herein, or could instead be a sprague clutch bearing or ball bearing.

(34) Although the electric linear actuators **10**, **110** shown and described herein are rod-type actuators, a one-way bearing and thrust bearing located in the same or a similar location within a track-type actuator would function in the same way to prevent loads applied to the spindle nut via the track from back-driving the motor when the motor is not provided with power.

(35) In the above description, certain terms have been used for brevity, clarity, and understanding. No unnecessary limitations are to be inferred therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed. The order of method steps or decisions shown in the Figures and described herein are not limiting on the appended claims unless logic would dictate otherwise. It should be understood that the decisions and steps can be undertaken in any logical order and/or simultaneously. The different systems and methods described herein may be used alone or in combination with other systems and methods. It is to be expected that various equivalents, alternatives and modifications are possible within the scope of the appended claims.

Claims

1. An electric linear actuator comprising: a housing extending along a longitudinal axis; a spindle provided in the housing and extending along the longitudinal axis from a proximal end to a distal end; a motor coupled to the proximal end of the spindle and configured to rotate the spindle; a rod coaxially coupled to the spindle and movable with respect to the housing along the longitudinal axis in response to rotation of the spindle; a direct drive coupling configured to couple the proximal end of the spindle to an output shaft of the motor; a thrust bearing provided coaxially on the spindle, wherein the thrust bearing is located between the proximal end of the spindle and the rod; and a one-way bearing provided coaxially on the spindle and configured to prevent back-drive of the motor when the spindle is subject to an external load via the rod, wherein the one-way bearing is located adjacent the proximal end of the spindle between the proximal end of the spindle and the thrust bearing, and between the direct drive coupling and the thrust bearing, and wherein the one-way bearing comprises: a first side; and a second side; wherein the one-way bearing is configured to: facilitate free rotation between a first bearing surface located on the first side and a second bearing surface located on the second side in a first direction; and inhibit rotation between the first bearing surface and the second bearing surface in a second direction opposite the first direction; wherein the one-way bearing is located within the housing such that one of the first and second sides is inhibited from rotating within the housing via a coupling to an inside of the housing, and the other of the first and second sides is coupled to the spindle, thereby facilitating rotation of the spindle in the first direction, and inhibiting rotation of the spindle in the second direction while the motor is not provided with electrical power.
2. The electric linear actuator of claim 1, further comprising a third bearing surface located on a proximal side of the thrust bearing and a fourth bearing surface located on a distal side of the thrust bearing.
3. The electric linear actuator of claim 2, wherein one of the third and fourth bearing surfaces is provided on a thrust race that is provided coaxially on the spindle, and the other of the third and fourth bearing surfaces is provided on a thrust washer.
4. The electric linear actuator of claim 1, wherein one of the first and second bearing surfaces is provided on a thrust race that is provided coaxially on the spindle; and wherein the thrust race also serves as an inner race for the one-way bearing.
5. The electric linear actuator of claim 4, wherein the spindle is configured to rotate within the thrust race as the motor rotates the spindle in the second direction.

6. The electric linear actuator of claim 4, wherein the thrust race comprises a hollow cylindrical portion serving as the inner race for the one-way bearing and a flange extending radially outwardly from the cylindrical portion, the flange comprising a bearing surface for the thrust bearing.
7. The electric linear actuator of claim 1, wherein the one-way bearing is a roller bearing having rollers with axes of rotation parallel to the longitudinal axis.
8. The electric linear actuator of claim 1, further comprising: a thrust race provided coaxially on the spindle, the thrust race comprising: a hollow cylindrical portion; and a flange extending radially outward from the cylindrical portion, wherein the thrust race comprises both a radially oriented bearing surface for the one-way bearing and a longitudinally oriented bearing surface for the thrust bearing.
9. The electric linear actuator claim 8, wherein the hollow cylindrical portion comprises the radially oriented bearing surface and the flange comprises the longitudinally oriented bearing surface.
10. The electric linear actuator of claim 1, further comprising: a thrust race provided coaxially on the spindle, wherein the thrust race serves as an inner race of the one-way bearing, thereby providing one of the one of the first and second bearing surfaces, wherein the spindle is configured to rotate within the thrust race in the second direction as the motor rotates the spindle in the second direction, and wherein the one-way bearing is configured to inhibit rotation of the thrust race in the second direction as the motor rotates the spindle in the second direction, thereby facilitating rotation of the spindle in the second direction with respect to the thrust race and the housing.
11. An electric linear actuator comprising: a housing extending along a longitudinal axis, the housing configured to be coupled to a first component; a spindle provided in the housing and extending along the longitudinal axis from a proximal end to a distal end; a motor coupled to the proximal end of the spindle and configured to rotate the spindle; a spindle nut received on the spindle and movable with respect to the housing along the longitudinal axis in response to rotation of the spindle, the spindle nut configured to be coupled to a second component; a thrust race provided coaxially on the spindle, the thrust race comprising: a hollow cylindrical portion; and a flange extending radially outward from the cylindrical portion; a one-way bearing configured to prevent back-drive of the motor when an external load is applied to the second component while the motor is not provided with electrical power, wherein the one-way bearing is provided coaxially to the spindle and the thrust race, and is located between the hollow cylindrical portion of the thrust race and the housing; and a thrust bearing provided coaxially on the spindle, wherein the thrust race comprises both a radially oriented bearing surface for the one-way bearing and a longitudinally oriented bearing surface for the thrust bearing.
12. The electric linear actuator of claim 11, wherein the hollow cylindrical portion comprises the radially oriented bearing surface and the flange comprises the longitudinally oriented bearing surface.
13. The electric linear actuator of claim 11, wherein the one-way bearing is configured to allow the thrust race and the spindle to rotate freely in a high-efficiency direction of the electric linear actuator configured to actuate the electric linear actuator in a first linear direction, such that the thrust race and spindle both rotate in the high efficiency direction during actuation of the electric linear actuator in the first linear direction; and wherein the spindle is configured to rotate within the thrust race as the motor rotates the spindle in an opposite, low-efficiency direction configured to actuate the electric linear actuator in a second linear direction, such that the spindle rotates within the thrust race in the low-efficiency direction while the thrust race is inhibited from rotating in the low-efficiency direction with respect to the housing during actuation of the electric linear actuator in the second linear direction.
14. The electric linear actuator of claim 11, wherein the spindle is coupled to the motor via a mechanical coupling member provided coaxially adjacent the proximal end of the spindle; and wherein the one-way bearing is located adjacent the proximal end of the spindle, and between the mechanical coupling member and the spindle nut.

15. The electric linear actuator of claim 14, wherein the one-way bearing is located between the proximal end of the spindle and the thrust bearing.
 16. The electric linear actuator of claim 11, wherein the one-way bearing is a roller bearing having rollers with axes of rotation parallel to the longitudinal axis.
 17. The electric linear actuator of claim 11, further comprising a rod configured to couple the spindle nut to the second component and configured to extend and retract from the housing as the spindle nut travels along the spindle.
 18. The electric linear actuator of claim 11, wherein the one-way bearing comprises an outer race; wherein an outer surface of the hollow cylindrical portion serves as an inner race for the one-way bearing; and wherein the outer race is adjacent to, and in contact with, a portion of the housing.
 19. The electric linear actuator of claim 11, further comprising: a direct drive coupling configured to couple the proximal end of the spindle to an output shaft of the motor, wherein the one-way bearing is located adjacent the proximal end of the spindle between the proximal end of the spindle and the thrust bearing, and between the direct drive coupling and the thrust bearing.
 20. The electric linear actuator of claim 11, further comprising: a bushing located between the housing and the one-way bearing, wherein the bushing serves as a housing for the one-way bearing; and a gear provided coaxially on the spindle, and comprising a portion of a gear chain coupling configured to couple the proximal end of the spindle to the motor.
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